

Twenty-five conjectures from a local–global random model, with machine verification and adversarial audit

The Conjecture Engine*

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Abstract

We state twenty-five conjectures in elementary and analytic number theory, each derived from the local–global random model of the primes: Cramér’s model corrected by Hardy–Littlewood singular series, with Bateman–Horn as the organizing framework and Borel–Cantelli accounting for sparse events. Each statement is calibrated to the strongest form the heuristic supports and no stronger. Every computable constant is computed from its definition, with local admissibility asserted programmatically; every statement was tested against exact counts, with success measured by the *shape* of agreement (predicted constant, square-root residuals, no drift) rather than by the size of the range searched. The statements were then subjected to a three-layer adversarial audit: an independent literature search for prior art, scripted refutation attempts a decade beyond the original bounds, and clean-context re-derivations of constants and counts. The audit refuted one conjecture as first stated—the exceptional set for $n = p + k^3$ contains the infinite family of cubes k^3 with $3k^2 - 3k + 1$ composite, an algebraic obstruction the probabilistic accounting missed—and it appears here restated, with the obstruction promoted to a theorem and the original error documented. Fifteen of the twenty-five turn out to be previously stated in essence and are attributed; ten carry instance-level novelty. The most substantial new claims are: a quantified Chebyshev-type race among twin-prime pairs modulo 5 whose entire bias is driven by the pairs $(q^2 - 2, q^2)$; a finite-modulus deficit law for least primes in arithmetic progressions, with a reproducible anomaly showing the ordered sequence of primes fills residue classes faster than any exchangeable model; and a Granville-corrected first-occurrence law for prime gaps with slope $\sqrt{e^\gamma/2} = 0.943682\dots$, registered against Shanks’s classical constant 1. A fourth statement—a finite- x variance law for prime counts in microscopic windows with second-order constant $\gamma + \log 2\pi - 1$ —transports Montgomery–Soundararajan’s mesoscopic variance into Gallagher’s regime and is verified to three decimals; it has numerical precedent and is claimed only as a sharpened instance.

Contents

1	Introduction	2
2	Notation	3

*This paper was produced by an automated conjecture engine (Claude, Anthropic), operated as an exercise in disciplined conjecture-making. All constants, counts, and audit results are reproducible from the accompanying repository (**conjecture-engine**): `run_all.py` regenerates every number in this paper. Primality of integers beyond 3.3×10^{24} was tested with Baillie–PSW; all smaller primality claims are deterministic (fixed-base Miller–Rabin).

3	Bateman–Horn instances	3
4	Sparse sequences: barely-divergent Borel–Cantelli	5
5	Representation problems	6
6	Statistical laws	8
7	The adversarial audit	10
8	Attribution and novelty	11

1 Introduction

A good conjecture in the theory of numbers is not merely a statement that has resisted counterexample. It is the *prediction of a model*: the assertion that the primes, beyond their local structure, behave like a random set of density $1/\log n$. The discipline this imposes is exacting. A conjecture worth stating should (i) survive every congruence and size obstruction, which is a computation, not a hope; (ii) come with a quantitative asymptotic whose constant is derived, not fitted; (iii) demand exactly the fluctuations the model earns—square-root noise with logarithmic corrections—and no fewer (the Mertens conjecture died of demanding fewer); (iv) sit inside the standard hierarchy of conjectures (Hardy–Littlewood k -tuples $\subset$ Schinzel’s Hypothesis $H \subset$ Bateman–Horn), so that its failure would propagate; and (v) be falsifiable instance-by-instance by computation. Numerical range is the least important column in the ledger: most deep phenomena drift at the rate $\log \log x$, and $\log \log 10^{18} \approx 3.7$, so verification “to 10^{18} ” is weak evidence by itself. What persuades is the *shape* of the agreement—a count that tracks a derived constant through several decades with residuals that look like noise.

This paper is the output of an automated engine built to those specifications, together with the audit that the specifications demand. The engine’s twenty-five statements are organized in four blocks: Bateman–Horn systems (§3), barely-divergent sparse sequences (§4), representation problems with Borel–Cantelli accounting (§5), and statistical laws of the prime sequence (§6). Section 7 describes the adversarial audit: what was attacked, what survived, and in one case what did not. We consider the audit trail part of the mathematical content. Its headline is instructive: the one outright refutation (Conjecture 13, first version) was produced not by bad luck but by a density-zero algebraic family—the cubes—invisible to a probability model that integrates over density. This is precisely the classical lesson of admissibility, and the engine had to relearn it in public.

Two further audit outcomes deserve emphasis. First, an independent literature layer found that fifteen of the twenty-five statements already exist in essence—some verbatim (Dubner’s twin-sum conjecture, the Stern list for $p + 2k^2$, Caldwell–Gallot’s $e^\gamma \log N$ laws), one with our exact constant catalogued as an OEIS entry (A188596 for Conjecture 6). We keep these statements in the paper, restated in our uniform framework and re-verified at our bounds, but they are attributed explicitly in §8; the residual claim to novelty of this paper rests on the other ten. Second, several of the engine’s *first drafts* were corrected by data before any external audit: the naive Poisson claim behind Conjecture 20 fails at every accessible scale (the corrected second-order law is the conjecture); the first-guess bias direction in the twin race of Conjecture 21 was reversed by a mod-3 computation; the naive $\text{Exp}(1)$ mean in Conjecture 22 is off by a factor

that took three controls to localize. We report these corrections because the difference between a fitted claim and a derived claim is exactly what an audit is for.

Verification standards

All counts below are exact (sieves; deterministic Miller–Rabin for $n < 3.3 \times 10^{24}$; Baillie–PSW beyond, flagged as such). Every comparison reports the Poisson-normalized residual $z = (\text{obs} - \text{pred})/\sqrt{\text{pred}}$. For sparse counts the cumulative comparison is contaminated at small x by the phantom mass of the main-term integral near its lower limit; in those cases we also report per-decade increments, which are clean. Constants are computed by truncated Euler products with the truncation wobble (difference between cutoffs) reported as an error bar; for conditionally convergent quadratic and cubic products this wobble, not the last digit, is the honest precision.

2 Notation

p, q denote primes; γ is Euler’s constant, $e^\gamma = 1.781072\dots$; φ is Euler’s totient; $\phi = (1 + \sqrt{5})/2$; F_n the Fibonacci numbers; $p\#$ the primorial. For irreducible $f_1, \dots, f_k \in \mathbb{Z}[x]$ with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \dots, f_k) = \prod_p \frac{1 - \omega(p)/p}{(1 - 1/p)^k}, \quad \omega(p) = \#\{n \bmod p : p \mid f_1(n) \cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) = \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual $1/\prod_i \deg f_i$. The Bateman–Horn conjecture [2] asserts $\#\{n \leq N : \text{all } f_i(n) \text{ prime}\} \sim C \cdot I(N)$ whenever $C \neq 0$; every conjecture in §3 is an instance with its constant evaluated. We write $\text{Li}_k(x) = \int_2^x (\log t)^{-k} dt$. The twin singular series is $\mathfrak{S}(d) = 2C_2 \prod_{p|d, p>2} \frac{p-1}{p-2}$ for even d , with $2C_2 = 1.3203236\dots$; the Goldbach series $\mathfrak{S}(n)$ is the same product over odd $p \mid n$.

Admissibility ($\omega(p) < p$ for all p) is asserted programmatically during every constant computation. The check does real work: the natural-looking pair $\{n^2 + n + 1, n^2 + n + 3\}$ has $\omega(3) = 3$ and is rejected; the engine’s Conjecture 5 uses the nearest admissible companion instead.

Novelty labels, from the independent literature audit (§8): **(a)** previously stated in essence; **(b)** classical family, quantified instance apparently unstated.

3 Bateman–Horn instances

The eight conjectures of this block share one hypothesis-generating move: choose an admissible system of polynomials, compute its singular series, and state the count. The heuristic is uniform, so we give it once. Model the primality of the values $f_i(n)$ as independent events of probability $1/\log f_i(n)$, corrected by the factor $(1 - \omega(p)/p)/(1 - 1/p)^k$ at each prime p to account for the joint local behaviour; multiplying the corrections and integrating gives $C \cdot I(N)$.

Conjecture 1 (Quadratic twin pair; (a), cf. OEIS A080149, [27]). $\#\{n \leq N : n^2 + 1, n^2 + 3 \text{ both prime}\} \sim C_1 I(N)$, $C_1 = C(x^2 + 1, x^2 + 3) = 2.954014\dots$

Here n must be even and $n \not\equiv 0 \pmod{3}$; for $p \geq 5$, $\omega(p) = 2 + (\frac{-1}{p}) + (\frac{-3}{p}) \leq 4 < p$. The conjecture implies infinitely many twin pairs $(m+1, m+3)$ with m a perfect square—a de Polignac refinement along a quadratic sequence.

Conjecture 2 (Cubic shift; (b)). $\#\{n \leq N : n^3 + 2 \text{ prime}\} \sim C_2^* I(N)$, $C_2^* = C(x^3 + 2) = 1.298435 \pm 0.0003$.

For $p \equiv 2 \pmod{3}$ the cube map is a bijection mod p , so $\omega(p) = 1$ *exactly* and the local factor is $(1 - 1/p)^{-1}(1 - 1/p) = 1$: only $p \equiv 1 \pmod{3}$ (where $\omega \in \{0, 3\}$ according as -2 is a cubic residue) moves the product, which therefore converges only conditionally—hence the quoted wobble. No statement of Bunyakovsky type is proven for any cubic; the nearest theorem is Heath-Brown’s $x^3 + 2y^3$ [3], which is the partial-result neighbour that makes the single-variable statement a research programme rather than a monolith.

Conjecture 3 (Chernick chain; (a), Dubner [8]). $\#\{p \leq x : p, 2p-1, 3p-2 \text{ all prime}\} \sim C_3 I(x)$, $C_3 = C(x, 2x-1, 3x-2) = 2.858249 \dots$

The three primes form an arithmetic progression with common difference $p-1$, first instance $(3, 5, 7)$; for $p = 6m+1$ the product $p(2p-1)(3p-2) = (6m+1)(12m+1)(18m+1)$ is Chernick’s universal-form Carmichael number, so this conjecture quantifies the population from which the classical Carmichael constructions draw.

Conjecture 4 (A diameter-14 quintuplet; (b)). *The pattern $(0, 2, 6, 12, 14)$ is admissible, and* $\#\{n \leq x : n, n+2, n+6, n+12, n+14 \text{ all prime}\} \sim C_4 \text{Li}_5(x)$, $C_4 = 15.19770 \dots$

The pattern differs from the two classical diameter-12 quintuplets $(0, 2, 6, 8, 12)$ and $(0, 4, 6, 10, 12)$: it contains two twin pairs separated by a gap of 4 around a middle prime. Its residues mod 5 are $\{0, 1, 2, 4\}$, leaving the class 3 free.

Conjecture 5 (Shifted quadratic pair; (b); no prior trace found). $\#\{n \leq N : n^2 + n + 1, n^2 + n + 7 \text{ both prime}\} \sim C_5 I(N)$, $C_5 = 5.928477 \dots$

Since $n^2 + n$ is always even, both forms are odd for every n : the local factor at 2 is $(1 - 0/2)/(1 - 1/2)^2 = 4$, a parity *bonus* rather than the usual halving. The nearer companion $\{n^2 + n + 1, n^2 + n + 3\}$ is inadmissible at $p = 3$, as noted above; $+7$ is the nearest shift that survives.

Conjecture 6 (Cyclotomic Sophie Germain; (a); the constant is OEIS A188596). $\#\{p \leq x : p^2 + p + 1 \text{ prime}\} \sim C_6 I(x)$, $C_6 = C(x, x^2 + x + 1) = 1.521661 \pm 0.0005$ (*catalogue value* $1.5217315 \dots$, *OEIS A188596*).

Since $p^2 + p + 1 = (p^3 - 1)/(p - 1)$ is the length-3 repunit in base p , the conjecture says: infinitely many prime bases have prime repunits of length 3. This is the base-independent home of the phenomenon whose base-2 shadow is the Mersenne sequence. Locally, $3 \mid p^2 + p + 1$ exactly when $p \equiv 1 \pmod{3}$, so half of all primes are barred at 3; for $q \equiv 1 \pmod{3}$ the two primitive cube roots of unity give $\omega(q) = 3$ (including the root from the linear factor).

Conjecture 7 (Downward companion; (b); sequence is OEIS A062326). $\#\{p \leq x : p^2 - 2 \text{ prime}\} \sim C_7 I(x)$, $C_7 = C(x, x^2 - 2) = 3.383216 \dots$

Squares are $\equiv 0, 1 \pmod{3}$, so $3 \nmid p^2 - 2$ always—no loss at 3—and $\omega(q) = 1 + (\frac{2}{q})$ for the quadratic factor. The same pairs $(p, p^2 - 2)$ reappear as the complete bias mechanism of the twin race in Conjecture 21; the two statements stand or fall together, which is the web-of-implications property one wants.

Conjecture 8 (Quadratic triple; (b); no prior trace found). $\#\{n \leq N : n^2 + 1, n^2 + 3, n^2 + 7 \text{ all prime}\} \sim C_8 I(N)$, $C_8 = 10.64599 \pm 0.006$.

This refines Conjecture 1 exactly as quadruplets refine twin primes, one level up from the linear world. Mod 7: $n^2 + 7$ bars $n \equiv 0$, $n^2 + 3$ bars $n \equiv \pm 2$, and $n^2 + 1$ has no roots, so $\omega(7) = 3$. The greedy extension by $n^2 + 9$ remains admissible but forces $n \equiv 0 \pmod{5}$, collapsing the candidate set to one lane; the triple is the natural stopping point.

Verification

	bound	observed	predicted	ratio	z
C1	10^7	32,898	32,862.7	1.0011	+0.19
C2	10^7	287,956	287,784.8	1.0006	+0.32
C3	$3 \cdot 10^8$	125,379	125,429.4	0.9996	−0.14
C4	$4 \cdot 10^9$	15,236	15,230.3	1.0004	+0.05
C5	10^7	65,871	65,953.6	0.9987	−0.32
C6	10^7	33,661	33,855.5	0.9943	−1.06
C7	10^7	74,914	75,273.8	0.9952	−1.31
C8	10^7	3,963	3,998.6	0.9911	−0.56

Independent clean-context recomputation gave $C_6, C_7, C_8 = 1.521730, 3.383227, 10.647706$ with primes to 10^9 (each inside our stated wobble) and independent counts to 3×10^7 with final ratios 0.9962, 0.9965, 1.0059. For the sparse pattern of C4 the cumulative ratio climbs $0.18 \rightarrow 0.59 \rightarrow 0.96 \rightarrow 1.0004$ between 10^4 and $4 \cdot 10^9$ while the per-decade increment ratio is already 1.00 ± 0.01 from 10^6 on: the early cumulative deficit is the phantom mass of Li_5 near $t = 2$, not a failure of the constant.

4 Sparse sequences: barely-divergent Borel–Cantelli

For sequences whose n th term has size e^{cn} the expected number of primes is $\sum_n \kappa_n/(cn)$: convergence or divergence of this sum is the entire question, and when it diverges it diverges logarithmically, so the right conjecture has counting function $\alpha \log x$ with a derived α —never more.

Conjecture 9 (Fibonacci primes; (a), cf. [5]). $\#\{p \leq x : F_p \text{ prime}\} \sim \frac{e^\gamma}{\log \phi} \log x = 3.70122 \dots \log x$.

Every prime factor of F_p ($p \neq 5$ prime) has rank of apparition exactly p , hence is $\equiv \pm 1 \pmod{p}$: the Lenstra–Pomerance–Wagstaff screening [6] transfers with $\log 2 \mapsto \log \phi$. The data sit persistently *below* the constant: 25 Fibonacci prime indices $\leq 10^4$ (independently reconfirmed by a second implementation) against a predicted 34.1, ratio ≈ 0.73 , $z = -1.6$, stable over three decades. Grantham and Granville [5] caution that naive constants for such sequences require $O(1)$ local corrections; the mod-5 divisor structure of F_p is the natural suspect, and we flag this constant as the component of the suite most likely to need a Granville-type patch.

Conjecture 10 (Primorial primes; (a), Caldwell–Gallot [4]). $\#\{p \leq x : p\# + 1 \text{ prime}\} \sim e^\gamma \log x$.

$p\# + 1$ is coprime to every prime $q \leq p$, a Mertens boost of $\prod_{q \leq p} (1 - 1/q)^{-1} \sim e^\gamma \log p$ against size $\log(p\#) = \theta(p) \sim p$: probability $e^\gamma \log p/p$ per prime, partial sums $e^\gamma \log x$.

Conjecture 11 ($n^2 + 2^n$; (b); sequence is OEIS A064539). *Every prime of the form $n^2 + 2^n$ with $n > 1$ has $n \equiv 3 \pmod{6}$, and*

$$\#\{n \leq N : n^2 + 2^n \text{ prime}\} \sim \sum_{\substack{n \leq N \\ n \equiv 3 \pmod{6}}} \frac{\kappa}{\log(n^2 + 2^n)}, \quad \kappa = \prod_p \frac{1 - \delta_p}{1 - 1/p} = 4.2734\dots,$$

where δ_p is the exact density of $n \equiv 3 \pmod{6}$ with $p \mid n^2 + 2^n$ over the period $\text{lcm}(6, p \cdot \text{ord}_p 2)$. In particular infinitely many such primes exist, with $\sim (\kappa/6 \log 2) \log N$ of them up to N .

The congruence claim is elementary: n odd is forced by parity, and for odd n with $3 \nmid n$ we have $n^2 + 2^n \equiv 1 + 2 \equiv 0 \pmod{3}$. What appears to be a harsh obstruction is, inside the surviving lane, a local bonus: $\delta_2 = \delta_3 = 0$ contribute the factors $2 \cdot \frac{3}{2} = 3$ to κ . Mixed polynomial–exponential sequences are exactly where hand-waving fails and the computed local product earns its keep.

Conjecture 12 (Factorial primes; (a), [4]). $\#\{n \leq N : n! + 1 \text{ prime}\} \sim e^\gamma \log N$.

Verification

C9: hits $\{3, 5, 7, \dots, 9311, 9677\}$, 25 of predicted 34.1 ($z = -1.56$; see the remark above). **C10**: 11 primes with $p \leq 4000$ against 14.8 ($z = -0.98$). **C11**: hits $\{3, 9, 15, 21, 33, 2007, 2127, 3759\}$ up to 4200: observed 8, model 8.19, $z = -0.07$; the lane constraint was verified exhaustively (no prime off $n \equiv 3 \pmod{6}$ to $n = 3000$, independently reconfirmed). **C12**: 15 primes with $n \leq 700$ against 11.7 ($z = +0.98$). The paired statements **C10/C12** share a constant but not a sequence, and the data straddle it ($z = -1.0$ and $+1.0$): that is what a correct shared model looks like.

5 Representation problems

Theorem 1 (Cube obstruction; found by the adversarial audit). *For $k \geq 2$, the cube k^3 is representable as $p + j^3$ with p prime, $j \geq 1$, if and only if $3k^2 - 3k + 1$ is prime. Consequently the set of integers not representable as $p + k^3$ is infinite: the non-representable cubes alone have counting function $\sim x^{1/3}$. (That the full exceptional set is $\sim x^{1/3}$ follows only when combined with Conjecture 13(i), and is conjectural.)*

Proof. $k^3 - j^3 = (k - j)(k^2 + kj + j^2)$; for $1 \leq j \leq k - 2$ both factors exceed 1, so the difference is composite. The only candidate is $j = k - 1$, giving $3k^2 - 3k + 1$. Since $3k^2 - 3k + 1$ is composite for a density-one set of k (all but $O(K/\log K)$ of $k \leq K$, by Chebyshev bounds for the quadratic’s prime values), all but a vanishing proportion of cubes are unrepresentable. $\square$

Finding 1. As first generated (“the exception set of $n = p + k^3$, $k \geq 1$, is finite”), the next conjecture was **false**, and the engine’s own adversarial battery refuted it: all 412 exceptions found in $(10^8, 10^9]$ are perfect cubes, explained by Theorem 1. The error is a textbook violation of the admissibility clause of the design criteria—an algebraic obstruction along a density-zero subsequence, invisible to a Borel–Cantelli sum taken over all n . We record the failure rather than hide it; the restated version separates theorem from conjecture.

Conjecture 13 (Prime plus a cube, restated; (a)-core: the non-cube finiteness is Hardy–Littlewood’s $E_3(X) = O(1)$ [1]). (i) *The set of non-cube integers $n \geq 2$ not representable as $p + k^3$ ($k \geq 1$) is finite; its per-decade census decays super-geometrically, and its largest element below 10^8 is 78,526,384.* (ii) *The representable cubes obey Bateman–Horn: $\#\{k \leq K : 3k^2 - 3k + 1 \text{ prime}\} \sim C(3x^2 - 3x + 1) I(K)$.*

Conjecture 14 (Stern list; (a): OEIS A060003). *Exactly ten odd integers are not of the form $p + 2k^2$ with $k \geq 1$:*

$$1, 3, 17, 137, 227, 977, 1187, 1493, 5777, 5993.$$

The expected number of representations of odd n is $\asymp \sqrt{n}/\log n$, so the expected number of exceptions beyond any X decays like $\sum_{n>X} \exp(-c\sqrt{n}/\log n)$: the list should be complete, and the seven prime members are precisely the Stern primes. Verified here to 10^9 , one decade past our original bound, with no eleventh exception.

Conjecture 15 (Goldbach in the (3, 3) lane; (a), K. Martin [9]). *Every $n \equiv 2 \pmod{4}$ with $n \geq 6$ is $p + q$ with $p \equiv q \equiv 3 \pmod{4}$, and the ordered representation count obeys*

$$R_3(n) \sim \frac{1}{2} \mathfrak{S}(n) \int_3^{n-3} \frac{dt}{\log t \log(n-t)}.$$

When $n \equiv 2 \pmod{4}$, parity forces $p \equiv q \pmod{4}$, so the representations split into a (1, 1) and a (3, 3) family of asymptotic ratio 1; the (3, 3) lane keeps its smallest representations ($3 + q$) and is the lane with *no* exceptions at all, which is why the uniform threshold $n \geq 6$ can be conjectured. Verified exhaustively to 10^9 ; the count formula matches to 0.2–1.3% at $n \approx 10^6, 10^7, 10^8$.

Conjecture 16 (Uniform quantitative de Polignac; (a)-core). $\pi_d(x) = \#\{p \leq x - d : p, p + d \text{ prime}\} = \mathfrak{S}(d) \text{Li}_2(x) (1 + o(1))$ *uniformly for even $d \leq (\log x)^A$; at fixed large x the residuals z_d over even $d \leq 2000$ form a single mean-zero, unit-variance-order Gaussian profile with regression slope 1 and no trend in d .*

The force of the statement is its uniformity: the singular series takes a thousand different values over $d \leq 2000$, spanning a factor of about three between $d = 2$ and the primorial-rich d , and the counts must reproduce the entire profile simultaneously with square-root errors. A conspiracy would need to bend a thousand-dimensional vector, not one number. At $x = 10^8$: correlation 0.999999, slope 0.99966, $\max_d |z_d| = 1.80$; stressed to $d \leq 6000$, $\max |z_d| = 2.01$ over three thousand values.

Conjecture 17 (Dubner’s conjecture; (a) [7]). *Every even $n \geq 4210$ is a sum of two members of twin-prime pairs; the 35 exceptions (OEIS A007534, largest 4208) are complete.*

Twin members have density $\asymp 1/\log^2 x$, so expected representations grow like $n/\log^4 n$ and exceptions must be finite; the observed exceptions cluster in triples of consecutive even numbers, a structural fingerprint of the sparse low-lying twin population. Verified to 10^8 , adversarially re-swept to 10^9 , and independently re-verified to 10^9 by a second implementation with a different algorithm: the same 35 exceptions each time.

6 Statistical laws

Conjecture 18 (Twin-gap records; (a), Kourbatov [11]). *The maximal gap $G_t(x)$ between consecutive twin-prime starts up to x satisfies $G_t(x) \asymp \log^3 x$, with*

$$\limsup_{x \rightarrow \infty} \frac{G_t(x)}{\log^3 x} = \frac{1}{2C_2} = 0.757390\dots$$

to first order.

Twin starts thin the integers to density $2C_2/\log^2 t$; a Cramér process at that density has record waiting times $\sim$ (mean spacing) $\times$ $\log$ (number of events) $= \log^3 x/2C_2$. The Cramér–Granville episode teaches that such constants can be shifted by local corrections (Granville’s $2e^{-\gamma}$ [13]); the claim to take seriously is the cube of the logarithm, and the constant is flagged as first-order. Records to $4 \cdot 10^9$ (largest observed twin-to-twin gap 5292, following the twin at 2,466,641,069) wander in $[0.41, 0.56] \cdot \log^3 x$, approaching the asymptote from below on a $\log \log$ clock, as such statistics must.

Conjecture 19 (First occurrences, Granville-corrected Shanks; (b)). *Let $p(g)$ be the prime starting the first occurrence of a prime gap of exactly g . Then*

$$\frac{\log p(g)}{\sqrt{g}} \longrightarrow \sqrt{e^\gamma/2} = 0.943682\dots,$$

approached from above.

Shanks’s heuristic [12] asserts the limit 1; Wolf’s refinement $p(g) \sim \sqrt{g} e^{\sqrt{g}}$ and the Kourbatov–Wolf first-occurrence laws in residue classes [21] keep slope 1. But if maximal gaps obey Granville’s $\limsup G(x)/\log^2 x = 2e^{-\gamma}$, then first occurrences must follow the corrected constant $\sqrt{1/(2e^{-\gamma})}$. The two values differ by 6%—beneath the resolution of any feasible computation (our regression slope over $g \geq 100$ falls from 1.307 at 10^9 to 1.297 at $4 \cdot 10^9$, still far above both candidates and decreasing, as it must for either). We note what the referee audit made explicit: the stated limit implies $g/\log^2 p(g) \rightarrow 2e^{-\gamma}$ along *all* large g —a liminf strengthening of Granville’s limsup claim, in tension with Firoozbakht-type bounds that current numerics weakly favour. The conjecture’s value is to register a definite competing constant on the books: the two laws are distinguishable in principle, not yet in silicon.

Conjecture 20 (Microscopic variance law; (b); the constant is Montgomery–Soundararajan’s [15]). *Fix $\lambda > 0$ and let X count primes in $[t, t + \lambda \log x)$ for t random near x . Then $X \Rightarrow \text{Poisson}(\lambda)$ (Gallagher [14]), and at finite x*

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + O\left(\frac{1}{\log^2 x}\right),$$

with $\gamma + \log 2\pi - 1 = 1.41509\dots$

The deficit is forced by Montgomery’s singular-series average $\sum_{d \leq h} \mathfrak{S}(d)(h - d) = \frac{h^2}{2} - \frac{h}{2}(\log h + \gamma + \log 2\pi - 1) + o(h)$ applied in the microscopic window $h = \lambda \log x$: the second factorial moment of X is $\lambda^2 - \lambda(\log h + \gamma + \log 2\pi - 1)/\log x + \dots$, and the variance deficit follows. The naive claim $\text{Var}/\mathbb{E} = 1$ fails at every accessible scale by 18–28%; the corrected law matches with no fitted parameter. (Numerical precedent: Sanchis-Lozano [22] fitted the Montgomery–Soundararajan variance shape at $x \sim 10^8$ in the $h \gg \log N$ regime; the microscopic $h = \lambda \log x$ statement and test appear to be new, and we claim no more than that.)

	$x = 10^9$		$x = 4 \cdot 10^9$	
λ	observed	predicted	observed	predicted
1/2	0.8241	0.8189	—	—
1	0.7910	0.7854	0.7998	0.7960
2	0.7527	0.7520	0.7670	0.7646
4	0.7201	0.7185	—	—

The residual $\approx +0.003$ is the anticipated $O(1/\log^2 x)$ term.

Conjecture 21 (Twin race modulo 5; (b), apparently new). *Twin starts $p > 5$ lie in classes $p \equiv 1, 2, 4 \pmod{5}$. Since $3 \mid q^2 + 2$ for every prime $q > 3$, the only pairs $(n, n+2)$ with one entry a prime square that contribute prime-power terms to the weighted count $\sum_{n \leq x} \Lambda(n) \Lambda(n+2)$ are $(q^2 - 2, q^2)$; and $q^2 - 2 \pmod{5} \in \{2, 4\}$, never 1. Consequently, with $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$: (i) class 1 carries a systematic surplus*

$$D_1^{\text{sys}}(x) = \frac{1}{2 \log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2),$$

whose scale is governed by the constant C_7 of Conjecture 7; (ii) classes 2 and 4 are symmetric; (iii) the bias is smaller than the random-walk noise $\sqrt{\pi_t(x)}$ by a factor $\log x$, so—in contrast to Chebyshev’s races—the twin race has no persistent leader: the logarithmic density of $\{D_1 > 0\}$ tends to a limit strictly between $\frac{1}{2}$ and 1.

The mechanism is the twin analogue of the classical explanation of Chebyshev’s bias: prime-square terms contaminate the classes that can contain them. (Biases of twin primes in residue classes have been studied before—Sahoo [23] for the mod-4 race, and [24] for prime pairs versus isolated primes—but we could find no prior statement of the mod-5 race, the $(q^2 - 2, q^2)$ mechanism, or the quantified surplus below.) Mod 3 kills the pattern $(q^2, q^2 + 2)$ outright, an amusing local accident that halves the mechanism and (as a mod-3 computation showed, overturning the engine’s first guess of a quadratic-residue bias against class 4) concentrates the surplus entirely on class 1. Data to $4 \cdot 10^9$: class counts (3980017, 3982505, 3981922), D_1 within one noise unit of D_1^{sys} throughout, no persistent leader, and the control race $\pi_{t,2} - \pi_{t,4}$ wandering like a fair coin—all three clauses as predicted, none provable at present.

Conjecture 22 (Least primes in progressions; (a) for part (i) [16, 17, 25], (b) for part (ii)). *Let $p(a, q)$ be the least prime $\equiv a \pmod{q}$ and $U(a, q) = \text{Li}(p(a, q))/\varphi(q)$. Then: (i) $U \Rightarrow \text{Exp}(1)$ as $q \rightarrow \infty$; in particular $\max_a U - H_{\varphi(q)}$ converges to a Gumbel law ($H_n = \sum_{k \leq n} 1/k$), which is the order-statistics form of Wagstaff’s $\max_a p(a, q) \sim \varphi(q) \log^2 q$; (ii) the finite- q deficit is of first order $1/\log q$:*

$$\mathbb{E}_a[U] = 1 - \frac{\theta(q)}{\log q} + o\left(\frac{1}{\log q}\right),$$

with θ slowly varying; measured, $\theta \approx 1.5\text{--}1.9$ for $q \in [10^2, 6 \times 10^3]$, with $\mathbb{E}_a[U]$ bottoming near 0.737 around $q \approx 200$ and recovering through 0.776 by $q \approx 6000$.

Remark 1. *The deficit is reproducible and is not an artifact of the time-change: a Cramér pseudo-prime control (independent Bernoulli “primes” with the correct densities and parity) shows roughly half of it—the discreteness of large per-candidate hazards—while the other half*

vanishes under any reshuffling of the actual prime residues (iid labels, or a random permutation of the true residue sequence, both restore $\mathbb{E}[U] = 1.00$). The ordered sequence of primes fills residue classes measurably faster than any exchangeable model. We measure θ ; we cannot yet derive it. It is the deepest unexplained number produced by this exercise.

Conjecture 23 (Fermat quotients and the Wieferich ledger; (a), cf. [18]). With $q_p = (2^{p-1} - 1)/p \bmod p$: (i) q_p/p equidistributes on $[0, 1)$ with square-root discrepancy (Kolmogorov–Smirnov distance $O(\pi(x)^{-1/2})$, no drift); (ii) $\#\{p \leq x : q_p < K\} \sim \sum_{p \leq x} \min(1, K/p)$; (iii) hence $\#\{\text{Wieferich } p \leq x\} = \log \log x + O(1)$: exactly $\{1093, 3511\}$ below 10^8 , with a third example expected only at doubly-exponential heights.

Clause (i) is what the data can actually measure well, and it is strictly stronger than the Wieferich count: a desert or glut violating (iii) would break (i) first. Measured $\sqrt{n}$ KS = 0.82, 0.82, 0.52, 0.63 at $x = 10^5, 10^6, 10^7, 10^8$ (bounded, no drift); small-quotient census at $K = 100$: observed 169 against model 161.2.

Conjecture 24 (Conjecture F as a family; (a), [1], cf. [19]). For odd A let $Q_A(N) = \#\{n \leq N : n^2 + n + A \text{ prime}\}$ and $C(A) = C(x^2 + x + A)$. Then $Q_A(N) = C(A) I_A(N) (1 + o(1))$ uniformly over odd $A \leq 199$, with jointly Poisson-size residuals: the computed constants predict the complete empirical ranking of the hundred polynomials. Euler’s $A = 41$ tops the family, $C(41) = 6.6395463\dots$ (Cohen’s high-precision value [26]).

At $N = 10^6$: mean ratio 1.0006, standard deviation 0.0027, $\max |z| = 1.92$, rank correlation 0.99988; for $A = 41$, observed 261,080 against predicted 261,017.6 ($z = +0.12$, using Cohen’s constant). A cautionary note from the referee audit belongs here: our own truncated Euler product for $C(41)$ at cutoff $2 \cdot 10^5$ reads 6.64092—wrong in the fourth significant decimal—because these conditionally convergent products drift at the 10^{-3} scale over practical cutoffs. Quoted digits for all quadratic-family constants in this paper are limited by the stated truncation wobble, not by the last printed digit. The entire 10^8 -bit primality dataset compresses to one product formula per A —the description-length virtue in its purest form.

Conjecture 25 (Consecutive primes mod 3; (a), Lemke Oliver–Soundararajan [20]). For consecutive primes $p, p' \leq x$ (both > 3), the same-residue fraction $s(x)$ satisfies

$$\frac{1}{2} - s(x) = (c + o(1)) \frac{\log \log x}{\log x}, \quad c \in (0, \infty),$$

with the deficit split symmetrically between the $(1, 1)$ and $(2, 2)$ patterns.

The repulsion is real but evanescent—vanishing at the precise rate $\log \log x / \log x$, unlike Chebyshev’s persistent bias. Measured $\hat{c} = 0.400, 0.375, 0.370$ at $x = 10^7, 10^9, 4 \cdot 10^9$: the slow drift is consistent with the $o(1)$ but visible, so we treat the constant as $c \in [0.30, 0.40]$ pending the secondary terms of [20]; the symmetry check gives $\#(1, 1)/\#(2, 2) = 0.99989$ at $4 \cdot 10^9$.

7 The adversarial audit

Each statement above was attacked three ways.

Layer 1: literature. Five independent search agents combed OEIS, arXiv, and the standard references for prior art, producing the novelty labels and the attribution section below. This

layer also falsified one historical claim in the engine’s working notes (the $k \geq 1$ Stern list of Conjecture 14 was described as partly new; it is OEIS A060003 verbatim, and 1493 is the largest known Stern prime); the notes were corrected on the record.

Layer 2: scripted refutation. Every falsifiable-by-instance statement was pushed at least a decade past its original bound: exception hunts across $(10^8, 10^9]$ for Conjectures 13, 14, 15, 17; the uniformity of Conjecture 16 stressed on $d \leq 6000$; the statistical laws re-tested at $4 \cdot 10^9$ (Conjectures 18–21, 25, and the quintuplet count of Conjecture 4); the recovery clause of Conjecture 22 tested on fresh moduli $q \in (3000, 6000]$. Outcome: one kill (Finding 1); no other statement moved.

Layer 3: clean-context replication. Independent agents, given only the bare statements and forbidden to read the engine’s code or notes, recomputed constants from definitions and recounted sequences with their own implementations. All recomputed constants landed inside the stated truncation error; all recounts agreed (one replication initially disagreed at $z \approx -22$, traced by the replicating agent itself to a bug in *its* sieve—every prime $n \leq Q$ had been struck out by its own modulus—after which it agreed at $z < 1.6$). The same layer reproduced the Fibonacci deficit of Remark after Conjecture 9 and the ordering anomaly of Remark 1, confirming both are properties of the primes rather than of our code.

We draw one methodological conclusion. The only fatal error among twenty-five statements was not statistical but *algebraic*: a factorization living on a density-zero subsequence. Probabilistic sanity checks, however extensive, integrate over density and cannot see such families; only structural attack surfaces (here: “test the claimed exception census on special subsequences”) find them. An audit that only re-runs the original verification at a larger bound would have declared the false version of Conjecture 13 sound.

8 Attribution and novelty

Previously stated in essence, with our contribution being the uniform quantification, computed constants, and re-verification: C1 (OEIS A080149; [27]); C3 [8]; C6 (constant catalogued as OEIS A188596); C9 [5]; C10, C12 [4]; C13-core [1]; C14 (OEIS A060003); C15 [9]; C16-core (uniform Hardy–Littlewood B, cf. [10]); C17 [7]; C18 [11]; C23 [18]; C24 [1, 19]; C25 [20].

Instance-level novelty (family classical, quantified statement not found in the literature): the constants and quantified forms of C2, C4, C5, C7, C8, C11; the statements C19 (against the slope-1 laws of Shanks, Wolf, and Kourbatov–Wolf [21]), C20 (as a microscopic sharpening of [15], with the mesoscopic numerics of [22] as precedent), C21 (against the twin-bias literature [23, 24]), and C22(ii). Among these we consider C21 and C22(ii)—the twin-race mechanism and the least-prime deficit law with its ordering anomaly—the paper’s most genuinely new content, and Question 1 its best export.

Question 1. Determine $\theta(q)$ of Conjecture 22: derive, from the Hardy–Littlewood correlations or otherwise, the first-order deficit of $\mathbb{E}_a \text{Li}(p(a, q))/\varphi(q)$ below its random-model value 1—equivalently, explain quantitatively why the primes in their natural order occupy residue classes faster than any exchangeable resampling of themselves.

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